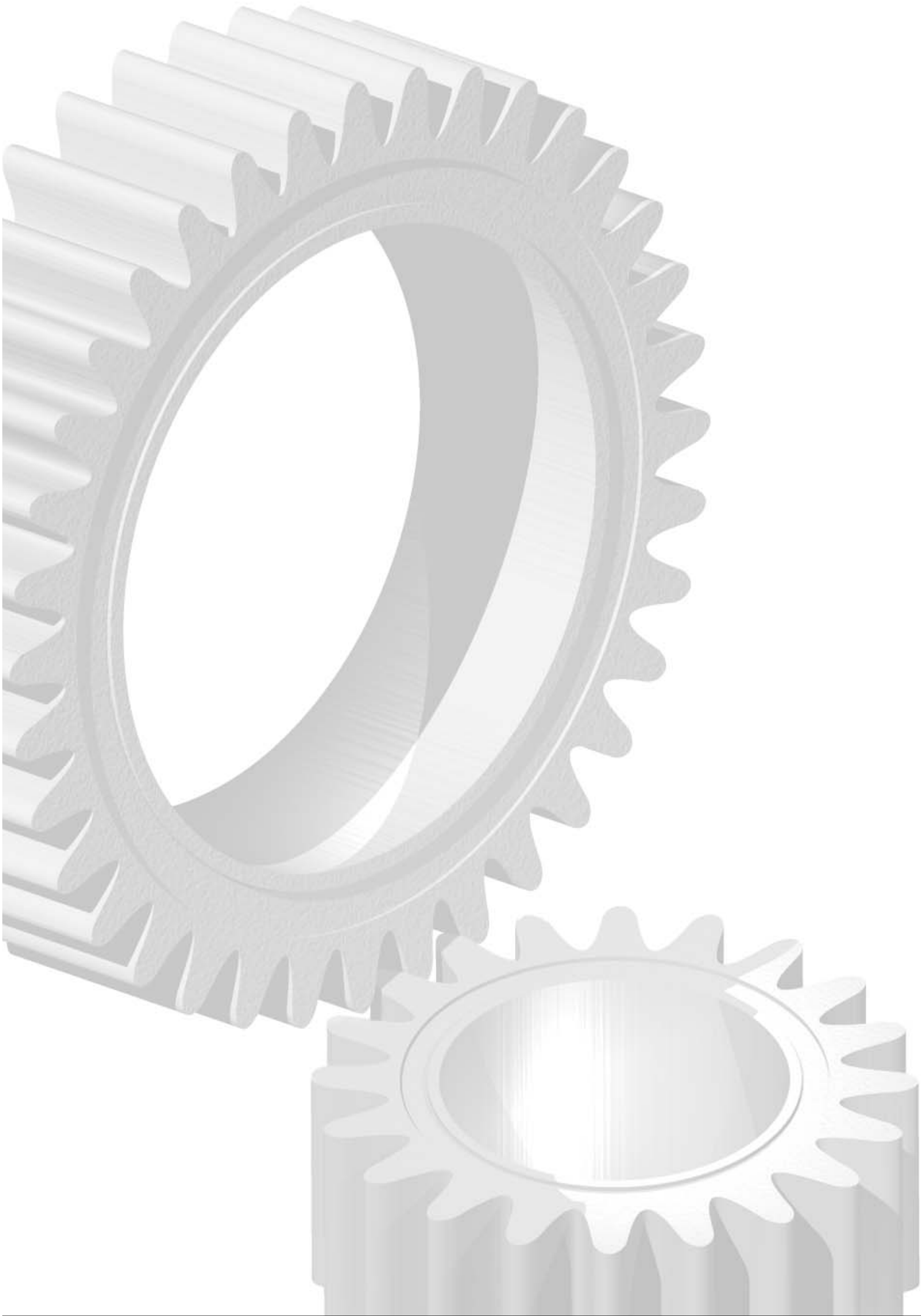
Technology White Paper



Autonomy

www.autonomy.com

Table of Contents

Autonomy Fundamentals

1.1 Introduction

In stark contrast to business practices of a few years ago, the modern enterprise is increasingly reliant on the efficient processing of unstructured information. Whether e-mails, Web pages or word processing documents, the content of unstructured information forms a critical link in virtually every value chain process across a wide range of business operations. The efficient management of such information is therefore directly linked to the bottom line. By automating key processes on unstructured information, Autonomy's technology enables the automation of business operations previously only performed manually. This offers significant savings for every type of organization and industry.

Analysts estimate that unstructured information is doubling in quantity every three months. Autonomy's ability to process all forms of unstructured information - including all formats on all platforms and even multiple streams such as audio and video - offers a unique solution to a growing number of applications and a host of platforms and devices that are increasingly dependent on utilizing unstructured information.

Autonomy employs a fundamentally different and unique combination of technologies to enable computers to form a contextual understanding of text, Web pages, e-mails, voice, documents and people's interests and hence, automate key operations upon them. Autonomy's technology is therefore able to power any application dependent upon unstructured information. This is evidenced by a significant penetration of the technology into a diversity of vertical markets, all of which need to manage and leverage the benefits of unstructured information.

Autonomy's unique combination of technologies, provides:

- **Accuracy**
- **Speed and performance**
- **Scalability**
- **Security**
- **Language Independence**
- **Easy integration**
- **Support for any content format**

Autonomy is therefore able to power any application dependent upon unstructured information including:

- *Business Intelligence*
- *Content Publishing*
- *E-Commerce*
- *Electronic Customer Relationship Management*
- *Email Routing*
- *Enterprise information portals*
- *Internet Portals*
- *Knowledge Management*
- *Online Publishing*

2. Intellectual foundations

Autonomy's strength lies in a unique combination of technologies that employs advanced pattern matching techniques (non-linear adaptive digital signal processing), utilizing Bayesian Inference and Claude Shannon's principles of information theory. Autonomy software identifies the patterns that naturally occur in text, based on the usage and frequency of words or terms that correspond to specific ideas or concepts. Based on the preponderance of one pattern over another in a piece of unstructured information, Autonomy enables computers to understand that there is X% probability that a document in question is about a specific subject. In this way, Autonomy is able to extract a document's digital essence, encode the unique "signature" of the concepts, then enable a host of operations to be performed on that text, automatically.

The innovative high-performance pattern-matching algorithms that provide the sophisticated contextual analysis and concept extraction, automate the categorization and cross-referencing of information, thereby improving the efficiency of information retrieval and enabling the dynamic personalization of digital content. For the first time, computers can now be enabled to automatically form an understanding of a page of text, Web pages, e-mails, voice, documents and people's interests, and automate operations dependant upon this unstructured content.

2.1 Bayesian inference

The theoretical underpinnings for Autonomy's approach can be traced back to Thomas Bayes, an 18th century English cleric whose works on mathematical probability were not published until after his death ("Philosophical Transactions of the Royal Society of London", 1763). Bayes' work centered on calculating the probabilistic relationship between multiple variables and determining the extent to which one variable impacts on another.

A typical problem is to judge how relevant a document is to a given query or agent profile. Bayesian theory aids in this calculation by relating this judgement to details that we already know, such as the model of an agent. More formally, the resulting, "a posteriori" distribution, which is applicable in judging relevance, can be given as a function of the known, "a priori" models and likelihood.

$$p(\theta|x) = \frac{p(x|\theta).p(\theta)}{\sum_{\theta' \in \Theta} p(x|\theta').p(\theta')}$$

Extensions of the theory go further than relevance information for a given query against a text. Adaptive Probabilistic Concept Modelling (APCM) analyzes the correlation between features found in documents relevant to an agent profile, finding new concepts and documents. Concepts important to sets of documents can be determined, allowing new documents to be accurately classified.

A traditional statistical argument is that if a coin is tossed 100 times and comes up heads every time, it still has an evens chance of coming up tails on the next throw. An alternative, Bayesian approach, is to say that 100 consecutive heads are evidence that the coin is not fair, or perhaps has heads on both sides. In a similar manner knowledge about the documents deemed relevant by a user to an agent's profile can be used in judging the relevance of future documents. APCM allows this information to be "back propagated"; in other words, agents can be improved by retraining.

Although no one knows for certain what Bayes' original goal was, Bayes' Theorem has become a central tenet of modern statistical probability modelling. By applying Contemporary computational power to the concepts pioneered by Bayes, it is now feasible to calculate the relationships between many variables quickly and efficiently, allowing software to manipulate concepts.

2.2 Shannon's information theory

Information Theory is the mathematical foundation for all digital communications systems.

Claude Shannon's innovation as described in his "Mathematical Theory of Communication" (1949) was to discover that "information" could be treated as a quantifiable value in communications.

Consider the basic case where the units of communication (for example, words or phrases) are independent of each other. If p_i is the probability of the i^{th} unit of communication, the average quantity of information conveyed by a unit, Shannon's entropy or measure of uncertainty is:

$$H=-\sum p_i.\log_2(p_i)$$

This formula reaches its maximum when the probabilities are all equal; in this case the resulting text would be random. If this is not the case the information conveyed by the text will be less than this maximum; in other words there is some redundancy. This result is then extended, by more sophisticated mathematical arguments, to when units are related.

Natural languages contain a high degree of redundancy. A conversation in a noisy room can be understood even when some of the words cannot be heard; the essence of a news article can be obtained by skimming over the text. Information theory provides a framework for extracting the concepts from the redundancy.

Autonomy's approach to concept modelling relies on Shannon's theory that the less frequently a unit of communication occurs, the more information it conveys. Therefore, ideas, which are more rare within the context of a communication, tend to be more indicative of its meaning. It is this theory that enables Autonomy's software to determine the most important (or informative) concepts within a document.

3 Infrastructure

3.1 Intelligent Data Operating Layer

Built upon this unique combination of technologies that employs advanced pattern matching techniques (non-linear adaptive digital signal processing), Autonomy delivers the Intelligent Data Operating Layer - IDOL, a powerful infrastructure technology which makes it possible for organizations to automatically process digital information.

IDOL forms an understanding of the content of any type of information, thereby enabling Integration Through Understanding (ITU), which allows applications to communicate with each other without any manual effort involved in setting up complicated connectors or the use of meta-data. Autonomy's technology makes enterprise systems "data-agnostic" and provides automated efficiencies never experienced before.

IDOL does not require complex programming, extensive integration, business rules, or middleware. It also does not require information to be manually tagged, linked or categorized. IDOL-compliant applications are immediately compatible through their common understanding of unstructured information. All this is possible because IDOL understands information in a way similar to humans - it directly relates concepts 'read' from the portions of documents humans can read, not from rules that are themselves dependent on synthetic tags.

More than 50 leading software companies are already incorporating Autonomy's infrastructure technology into the next version of their enterprise applications, whether for customer relationship management, e-business, customer care, e-mail routing and security, content delivery, or client-server systems. OEM partners delivering IDOL enabled applications include BEA, Vignette, Sybase, Computer Associates, Business Objects and Hyperwave.

3.1.1 The Dynamic Reasoning Engine™ (DRE™)

At the heart of Autonomy's IDOL is the Dynamic Reasoning Engine™ (DRE™), a scalable, multithreaded process that analyzes and delivers highly targeted content to users. The DRE is based on the advanced pattern-matching technology that exploits high-performance probabilistic modelling techniques. Autonomy's DRE performs the following core operations:

- **Concept matching:** The DRE™ accepts a piece of content* or reference (identifier) as an input and returns references to conceptually related documents ranked by relevance, or contextual distance. This is used to generate automatic hyper links between pieces of content.
- **Automatic Summarization:** The DRE™ accepts a piece of content and returns a summary of the information containing the most salient concepts in the content. In addition, summaries can be generated that relate to the context of the original inquiry - allowing the most applicable dynamic summary to be provided in the results of a given inquiry.
- **Active matching:** The DRE™ can accept textual information describing the current user task and returns a list of documents ordered by contextual relevance to the active task.

- **Automatic Hyperlinking:** The DRE™ dynamically links content* to contextually similar information, removing the requirement to manually insert hyperlinks to content
- **Contextual Retrieval:** The DRE™ accepts a Boolean term or natural language query and returns a list of documents containing the concepts looked for, ordered by contextual relevance to the query.

3.1.2 Classification Server™

In conjunction with the Dynamic Reasoning Engine's ability to understand any information contextually, Autonomy delivers IDOL's second key module "Classification Server™" that performs a wide range of highly scalable automatic classification solutions.

- **Automatic Clustering:** Classification Server™ organizes large volumes of content or large numbers of profiles into self-consistent clusters. Clustering is an automatic agglomerative technique, which partitions a corpus by grouping together information containing similar concepts.
- **Categorization:** Classification Server™ accepts a piece of content and returns categories ranked by conceptual similarity. This is used to discover which categories the content is most appropriate for, allowing subsequent tagging, routing or filing.
- **Automatic Taxonomy Generation:** Organizing large volumes of content or large numbers of profiles into self-consistent clusters enables the Classification Server™ to generate Taxonomies completely automatically. Clustering or any other conceptual operation can be used as a 'seed' to perform Automatic Taxonomy Generation.

3.1.3 User Agent Server™ (UA Server™)

Combining the Dynamic Reasoning Engine's with IDOL's third key module "User Agent Server," Autonomy delivers a range of powerful personalization operations including:

- **Agent creation:** The UA Server™ accepts a piece of content* and returns an encoded representation of the concepts, including each concept's specific underlying patterns of terms and associated probabilistic ratings.
- **Agent alerting:** The UA Server™ accepts a piece of content and returns similar agents ranked by conceptual similarity. This is used to discover users who are interested in the content, or find experts in a field.
- **Agent retraining**: The UA Server™ accepts an agent and a piece of content* and adapts the agent using the content.
- **Agent matching:** The UA Server™ accepts an agent and returns similar agents ranked by conceptual similarity. This is used to discover users with similar interests, or find experts in a field.

* 'Piece of content' refers to a sentence, paragraph or page of text, the body of an e-mail, a record containing human readable information, or the derived contextual information of an audio or speech extract.

4

Benefits of Autonomy's technology

4.1 Automation

Autonomy's technology facilitates a wide range of operations across all unstructured information formats. Given that such tasks have previously been performed by expensive, inaccurate and - inevitably slower - manual labor, Autonomy's technology offers a radically different automated solution for countless numbers of business operations and with it, a direct link to significant bottom line savings.

Virtually every market sector and industry has invested in Autonomy automating the processing and management of unstructured information. Given the uninterrupted growth of unstructured information, which is doubling every three months (Gartner), efficient processes that manage and extract value from such information are solely dependant on the ability to automate the tasks that previously, have been performed with manual labor.

4.2 Accuracy

Profiling and personalization are meaningless words to most people because most software applications use inaccurate key word, meta-tag or linguistic based technologies to attempt profiling and personalization emulation with, in most cases, hopeless and disappointingly coarse results. In contrast, Autonomy's technology provides highly accurate analysis of a user's information requirements utilizing both implicit and - where required - explicit techniques to establish dynamic, real-time results that go far beyond users experiences and in many cases, expectations.

4.3 Speed

Autonomy's technology is deployed to solve mission-critical business problems, with a wide variety of capacity and performance requirements. With ever-increasing volumes of users and data and operations taking place on a constant basis, the importance of immediate results and rapid and unpredictable increases in content and usage impose rigorous requirements on Autonomy's underlying technology. Real-world examples and benchmarks prove Autonomy surpasses most organizations' current load and performance expectations. Autonomy is already well prepared to handle the ongoing explosion of unstructured information.

4.4 Scalability

Because the Autonomy product architecture is designed to be completely modular, multithreaded and uses multiplexing sockets, Autonomy's infrastructure software provides a high performance, high capacity and scalable platform for content exploitation. It takes full advantage of massive parallelism, SMP processing environments and distributed server farms.

4.5 Security

Whether your company is implementing a knowledge management solution, publishing content, selling goods and services through electronic channels or providing application software products, a key consideration is security.

Autonomy's unique combination of sophisticated mathematical algorithms automates the processing and conceptual analysis of large volumes of both content and users, to provide highly personalized, content exploitation solutions without sacrificing critical security considerations.

Autonomy provides three basic forms of security. These are as follows:

1. **Authentication:** This governs who is able to log in to the system. Autonomy allows direct connections to the preferred authentication directory, such as Notes, NT, LDAP, Exchange, Netware or Oracle.

2. **Entitlement:** This governs which items in the results list can be seen by the user. In all corporate environments it is essential that underlying security entitlement models be respected. Autonomy remains synchronized with all underlying security models, including NT, Notes, Netware, Filenet, Documentum, Unix, Oracle/ODBC based sources, Exchange, etc. It does this efficiently, not just through callbacks to the repository for each result, which is a method that doesn't scale, and accurately. Updates and changes are immediately reflected in the Autonomy entitlement model.

3. **Authorization:** This governs who is able to view documents having clicked on the links in the results list and is not required with entitlement.

4.6 Language independence

It is critical for global companies today to be able to provide the right information to the right people, regardless of the language the content is represented in. Autonomy's technology is completely language independent. It does not rely on an intimate knowledge of English grammatical structure or that of any particular language. It treats words as abstract symbols of meaning, deriving its understanding through the context of their occurrence rather than a rigid definition of grammar. Autonomy's software supports over 56 languages, including English, German, French, Italian, Chinese, and Japanese. Autonomy's technology can even auto-detect the language of incoming documents and configure itself appropriately to deal with each.

4.7 Easy integration

Autonomy Application Builder™ and Active SDK™ have been designed to help companies and partners integrate Autonomy's technology to create their own customized applications. The provision of APIs in C, ActiveX, JAVA, PHP, EJB, TCL, COM+, C#, Perl and directly with HTTP means you can integrate with all professional development languages and Web scripting technologies.

4.8 Support for any format

Autonomy's technology aggregates content from any data source. Autonomy supports over 200 file formats, and can access repositories such as LotusNotes, Oracle, Exchange, etc.

5

5.0 Alternative approaches

Many companies claim to have solutions to the challenge of managing unstructured information, or have promised technologies to deliver personalized information services. However, most of these systems and approaches have severe limitations, particularly where scalability and cost are concerned. For example:

5.1 Keyword searching or Boolean query

The most common approach to information management is through traditional keyword search. This simple method involves asking a user to enter some terms into a text field. It then searches through a list of documents to return with a list of those containing the search terms.

5.1.1 Limitations

- **Reactive**
  Any solution requiring the user to realize they require some information, formulate a query and then perform that query is destined to only ever be a reactive solution. It will never be able to proactively alert a user to pertinent information, changes or emerging trends.
- **No context**
  The most common attempts to manage unstructured data employ keyword search. Using this method, a user enters a keyword or term into a text field, 'dog'

for example. The search engine then searches for documents that contain this term. The fundamental problem is that the search engine is simply looking for the occurrence of the search term in a document. This is regardless of how it is used or the context in which the user is interested in the word 'dog', rendering minimal the effectiveness of the information delivered.

- **Inaccurate**
  Reliance on inefficient computer linguistics, keyword definitions and tags produces inaccurate results and is normally costly to implement and maintain. Keyword searching is neither scalable nor deployable as a solution to the challenges that unstructured information presents in every market sector.
- **Manual**
  Keyword engines do nothing more complex than look for a few words, which is manually intensive at the back end, requiring humans to continually manage and update keyword associations or "topics". Such as dog=hound=canine or dog is 90% canine, 10% furry.
- **Intensive user participation**
  Keyword methodologies rely heavily on the sophistication of the end user to be able to author queries in fairly complex and specific language (also known as Boolean form) i.e. 'dog AND (NOT (Sleeping OR Rabid OR Running*) AND Sheep'. These technologies often rely on 'rich and powerful' query operators. Studies show that 99.9% of real life queries performed by end users consist of one or two keywords, and almost never contain sophisticated information retrieval operations. Inspection of your own log files will show similar trends.
- **Do not learn**
  Keyword search engines cannot "learn" through use, or be exposed to queries on the word dog and learn (without user intervention) that when dog is input, it is the four legged, furry kind of sheepdog for which information is sought. It is also very difficult for keyword search systems to find things by being shown an example. Typically, a "more like this..." function will simply increase the number of keywords in the query based on what terms appear most frequently in the example document. This will often result in more documents instead of less, which is what the user really wants.

5.1.2 Autonomy's approach

Autonomy's concept matching technology avoids these problems by matching concepts instead of simple keywords, although it does have the ability to perform legacy keyword and complex bracketed Boolean text queries as well. Autonomy takes into account the context in which terms appear. This eliminates many false hits while also catching documents that may not contain the specific term, but do include the concept. In addition it allows conceptual refinement, or refinement by example. Most users are able to point to one or more initial results and request more like those. Using the entire context of these selected pieces of content allows Autonomy to zoom in on the conceptual area that is really of interest.

5.2 Collaborative filtering or social agents

Collaborative Filtering is an attempt to allow computers to make personal recommendations to users based on their similarity to other users. The basic principle is quite simple: by getting a large number of users to give information about their preferences (usually by filling out forms and checking boxes) the system endeavors to make recommendations.

An example serves to clarify the basic principle. Imagine three users: Mick, Bud, and Brad have been asked to give their three favorite musicians.

Mick's favorite musicians:	*Elvis*	*Buddy Holly*	*Little Richard*
Bud's favorite musicians:	*Jimi Hendrix*	*James Brown*	*Aretha Franklin*
Brad's favorite musicians:	*Elvis*	*Jerry Lee Lewis*	*Little Richard*

In collaborative filtering the computer compares the results, finds that Mick and Brad are similar and so swaps each other's suggestions: "Mick, you may like Jerry Lee Lewis"; "Brad, you may like Buddy Holly".

5.2.1 Limitations

- **Limited use**
  Collaborative filtering only works well for a closed set of items (e.g. music, books). It does not work well for online services seeking to recommend relevant news stories or articles for purchase because the number of possible "subjects "is too large and diverse. It is also difficult to infer value from user rankings beyond the immediate set of items being ranked. Does indicating a preference for Elvis and Buddy Holly correlate with drinking Pepsi rather than Coke?
  This technique fails because it does not understand the content itself, but merely operates using statistics concerning usage, completely disregarding the content.
- **Not personalized**
  In the example above, the system assumes similar people act in similar ways. However, even for music, users tastes are complex. John may like Buddy Holly and Jerry Lee Lewis, hate Elvis, but like Mozart. The technology cannot take personal idiosyncrasies into account.
- **Intensive user participation**
  Collaborative filtering also requires tedious active participation from users. They must continuously fill out questionnaires or set ratings and grades for each object, a process which soon loses its novelty value.
- **Not scalable**
  This approach has inherent scalability problems for large numbers of users due to the multidimensional comparisons over all users. It is for these reasons that Forrester Research concluded in its May 1997 report "Personalize or Perish" that

collaborative filtering is useful in only a very small percentage of applications.

- **Cannot handle new information**
  The day one problem: On day one of a service there are no questionnaires for the system to work from. Collaborative filtering cannot deal very well with new information. In an E-Commerce environment, the system cannot recommend users who would be interested in a new product because nobody has purchased it before. For example, when the Spice Girls first came to prominence there were no existing questionnaires mentioning them.

5.2.2 Autonomy's approach

Autonomy's technology understands users' interests or employee expertise by extracting key ideas from the actual information the user reads. It then builds a profile that can be used for personalizing information or serve targeted advertising messages. Because these services are based on a user's actual interests, they do not need to fill out lengthy questionnaires or rate their likes and dislikes. These profiles can be kept completely anonymous and do not require the user to provide any private demographic information. However, profiles can be combined with any known demographic information to further personalize the services provided.

As an individual reads additional articles online, publishes material on the corporate Intranet or submits documents to the knowledge management system, the system updates the profile by recalculating interest levels in the different ideas. Concepts that once occurred frequently but no longer are important are replaced over time. In this way, the system keeps pace with an individual's changing interests. This is in contrast to an explicit preference setting, which users must remember to adjust as their interests evolve.

5.3 Parsing and natural language analysis

For the last twenty years much effort has been put into an obvious approach to deal with unstructured information called parsing (also semantic or lexical analysis). Rules of grammar and lexicons are applied to try to explicitly understand textual information.

Example:

The cat sat patiently on the mat. = (The cat = subject) (sat = verb) (patiently = adverb) (on = preposition) (the mat = object).

5.3.1 Limitations

- **No context**

  In spite of more than 20 years of research into parsing approaches, parsing is rarely used in real applications because of its poor performance for real-world problems. The following cases illustrate the limitations of this approach, namely, parsing's inability to handle ambiguity.

  Example 1:
  "The dog came into the room, it was white."

  It is unclear from the sentence whether it is the room or the dog that is white. On the other hand, a human being would have little problem deciphering the following examples because of his or her familiarity with both rooms and dogs:

  "The dog came into the room, it was furry."
  "The dog came into the room, it was full of furniture."

  The computer, however, would still be stumped. It lacks the understanding to solve such ambiguities. Some advanced systems will allow the construction of a set of rules for the machine to follow to resolve these uncertainties. However, the instruction set would be incredibly cumbersome and difficult to maintain, and would significantly degrade the system's performance.

  Example 2:
  "The fly, it's clear to me, can fly faster than the bee."

  First, the computer may be confused by the word "fly," which is used in this sentence as both a subject and a verb. But that is an easy problem to solve. What about the word "it"? How does one parse a word that refers to abstract thought? These problems are exacerbated when a computer attempts to extract meaning by parsing full paragraphs.

  Example 3:
  "The president arrived by car to meet the Chinese premier."

  Like keyword-based approaches, parsing cannot determine the relative importance of ideas. In other words, the computer will assign an equal level of importance to the President, his mode of transportation and the leader he is meeting with. Parsing at best can only handle a few sentences. A strict parsing mechanism has great difficulty extracting meaning from a full paragraph.
- **Unreliable**

  Because parsing is based on a true/false decision-tree structure, one incorrect decision can derail the entire analysis.

- **Language dependent**
  The parsing approach is language specific, and its reliance on the grammar of a given language means that it is vulnerable to slang or grammatically incorrect constructions. Because linguistic approaches base their understanding of 'foreign' content using a thesaurus, they cannot scale easily. This makes the deployment of multi-language or region specific applications and offerings very complicated and time-consuming.
- **Manual**
  Another approach championed by search vendors involves the linguistic processing of a question or command such as "Open Microsoft Word" or "Where will I find this?" While this can be appropriate for one sentence questions, or questions which concern a known universe of information, the language model simply breaks down when employed on large documents containing a large number of concepts. It is recognized, however, that this approach is useful to the market but it is a smaller market by definition and requires a significant amount of manual labor at the back-end.

5.3.2 Autonomy's approach

Autonomy's software avoids these problems because its pattern-matching technology uses predictable statistical word patterns to represent concepts and functions independently of any given language. The system is driven by the actual data fed into it, not by and auxiliary set of rules that are disconnected from the content.

5.4 Manual tagging

With an upswing in enterprise portals, creating taxonomies that address various information types (including documents, structured data, HTML, XML, and multimedia) is imperative. Manual tagging schemes are becoming an increasingly popular method of labelling digital material. However, cost is a significant barrier to ensuring that they increase the efficiency of managing information.

5.4.1 Limitations

- **Descriptive inconsistency**
  One example of the effect of human behavior and the inherent limitations of manually describing information - albeit from existing descriptions - is illustrated by the results of a US Department of Defense edict, mandating that internal users responsible for authoring documents also create an appropriate description of the content of the document. At first glance, a seemingly sensible and pragmatic decision. However, after many months of activity, it was discovered

that the vast majority of documents had been loosely described and tagged as 'general'. Whilst tagging schemes, and particularly XML, attempt to break away from such generalist terms, they remain dependant upon the same shortcomings of human behavior that manifest themselves as 'inconsistency'. An individual's ability to describe information is dependant upon their personal experience, knowledge and opinions. Such 'intangibles' vary from person to person and are also dependant upon circumstance, dramatically reducing the effectiveness of the results.

Further complications arise when subjects incorporate multiple themes. Should an article about 'technology development in Russia within the context of changing foreign policy' be classified as (i) Russian technology (ii) Russian foreign policy, or (iii) Russian economics? The decision process is both complex and time consuming and introduces yet more inconsistency, particularly when the sheer number of options available to a user is considered. For example, over 800 tags for general newspaper subjects make the task of choosing a potentially basic subject description in a reasonable time-scale an even more challenging process.

- **Idea Distancing**
  Tags also fail to highlight the relationships between subjects. Termed 'idea distancing', there are often vital relationships between seemingly separately tagged subjects such as wing design/low drag/and /aerofoil/efficiency/. The first category may contain information about the way the wings are designed to achieve low air resistance. The latter category discusses ways in which efficient aerofoils are made. Obviously, there will be a degree of overlap between these categories and because of this; a user may be interested in the contents of both. However, without understanding the meanings of the category names, there is no clear correlation between the two.
- **Not scalable**
  The organization aims to have complete control over its knowledge base, but in reality companies are overwhelmed with data. Unstructured information is growing exponentially, semi-structured and structured data are labor intensive and far too time consuming to process. The manual tagging solution to controlling this dilemma is defocused and doesn't scale.
- **High labor costs**
  Taxonomy creation and tagging is still a predominantly manual effort requiring input from librarians, users, and IT staff. This means large labor costs involved in making sense of information.
- **Specificity**
  In order to be highly specific in the retrieval and processing of tagged documents, the number of tags will need to be very high. For example, tag numbers in a company such as Reuters run into the tens of thousands. However, as the number of tags increases, so do both the effort involved and the likelihood of misclassification.

5.4.2 Autonomy's approach

Autonomy addresses the inefficiencies introduced by the manual issues associated with creating tags by adding a layer of intelligence to the management of XML: understanding the content and purpose of either the tag itself, or related information, or both.

Conclusion

6

Autonomy's technology provides a scalable and cost-effective alternative to existing approaches in managing unstructured data. Autonomy's technology is far superior to other approaches to handling unstructured information, such as keyword searches, which are highly inaccurate, and manual tagging linking or sorting of data, which is time, labour and cost-intensive.

Computers need to be smarter. They need to understand more about the information being communicated. We need to advance beyond worrying about where and how information is stored. Computers simply need to understand what information is stored - the rest is easy.

Autonomy looks forward to continuing progress in enabling computers to better understand not just the format of information, but also its meaning.

7 Further reading

- "Interactive Technology Strategies - Personalize or Perish?" The Forrester Report; Volume Two, Number Three, May 1997
- "Bayesian Methods Applied to Neural Networks" by Radford Neal, Springer Verlag
- "Bayesian Forecasting" by Mike West, Springer Verlag
- "Applied Bayesian Forecasting and Times Series Analysis" by Pole, West & Harrison, Chapman & Hall
- "Bayesian Statistics" by P. Lee, John Wiley & Sons
- "Numerical Bayesian Methods Applied to Signal Processing" by O-Ruinadah and W.J. Fitzgerald, Springer Verlag

Headquarters

Autonomy Inc.
301 Howard Street
San Francisco, CA 94105
Tel: 1 415 243 9955
Fax: 1 415 243 9984
Email: info@us.autonomy.com

Autonomy Systems Ltd
Cambridge Business Park
Cowley Road
Cambridge CB4 0WZ
United Kingdom
Tel: +44 (0) 1223 448 000
Fax: +44 (0) 1223 448 001
Email: autonomy@autonomy.com

Regional Offices

Autonomy Asia-Pacific
Level 14
33 Berry Street
North Sydney
NSW 2060
Australia
Tel: +61 (0) 2 9959 1951
Fax: +61 (0) 2 9959 1035
Email: australia@autonomy.com

Autonomy Belgium
Bessenveldstraat 25
1831 Diegem
Tel: +32 (0) 2 716 40 05
Fax: +32 (0) 2 716 41 92
Email: belgium@autonomy.com

Autonomy France
121, Avenue des Champs Elysées
75008 Paris
Tel: +33 (0) 1 53 57 29 92
Fax: +33 (0) 1 53 57 29 93
Email: france@autonomy.com

Autonomy Germany
Autonomy Germany - Frankfurt
Platz der Einheit 1
60327 Frankfurt am Main
Tel: +49 (0) 69 97503 246
Fax: +49 (0) 69 97503 336
Email: germany@autonomy.com

Autonomy Germany - Hamburg
Valentingskamp 24
20354 Hamburg
Tel: +49 (0) 40 31112 308
Fax: +49 (0) 40 31112 641
Email: germany@autonomy.com

Autonomy Germany - Munich
Arnulfstrasse 27
80335 Munchen
Tel: +49 (0) 89 59047 120
Fax: +49 (0) 89 59047 207
Email: germany@autonomy.com

Autonomy Italy
Largo Richini, 6
20122 Milano
Tel: +39 (0) 2 5821 5510
Fax: +39 (0) 2 5821 5400
Email: italy@autonomy.com

Autonomy Netherlands
Teleport Towers
Kingsfordweg 151
1043 GR Amsterdam
Postbus 57674
1040 BN Amsterdam
Tel: +31 (0) 20 491 9680
Fax: +31 (0) 20 491 7366
E-mail: netherlands@autonomy.com

Autonomy Nordic AS
P.O. Box 150
N-1324 Lysaker
Tel: +47 (0) 67 83 00 60
Fax: +47 (0) 67 83 00 61
Email: info@autonomy.no

Autonomy Spain / Conda.net
Grupo Afina
Hermosilla, 77-2°-1°
28001 Madrid
Tel: +34 (0) 91 781 24 56
Fax: +34 (0) 91 781 24 57
Email: spain@autonomy.com

Autonomy Sweden AB
Kungsgatan 48
11135 Stockholm
Tel: +46 (0) 8411 3550
Fax: +46 (0) 8411 3560
Email: swedishoffice@autonomy.com

(Autonomy Inc. and Autonomy Systems Limited are both subsidiaries of Autonomy Corporation plc)

[WP TECH] 07.02

www.autonomy.com

Autonomy ®